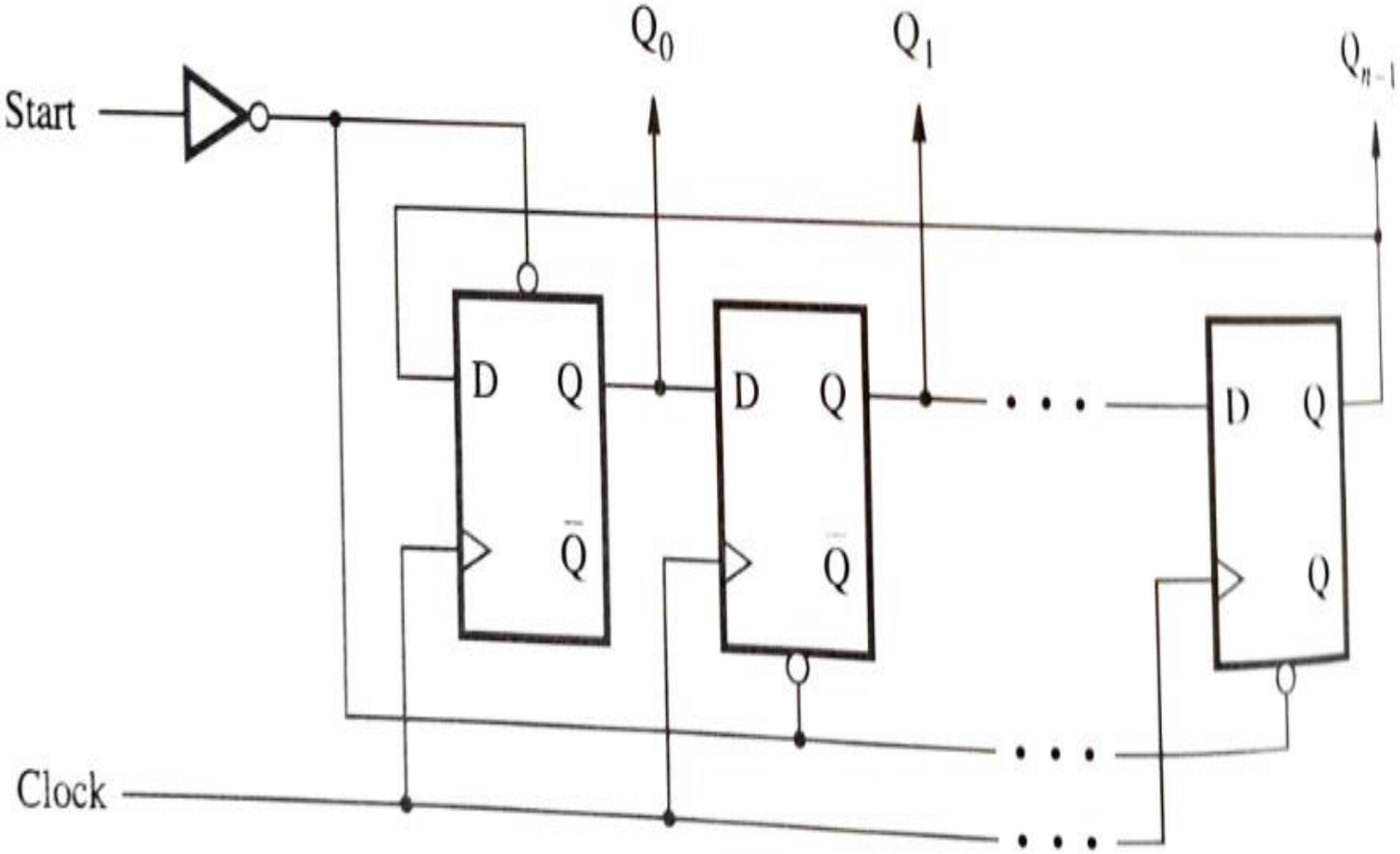
# Sequential Logic – Part 6
# K. Sridharan

# Review

- We have looked at a few types of counters.
- We will look at some more counters in this part.

# What does this circuit do ?



# Student exercise

- Hints: The Start line is connected (via an inverter) to the Preset input of the leftmost D flip-flop. Recall the Preset line in earlier discussion.
- The Start line is also connected (via an inverter) to the Clear input of the other flip-flops. Once again recall the purpose of the Clear input from earlier discussion.

# Class Exercise

- Determine how the circuit behaves.

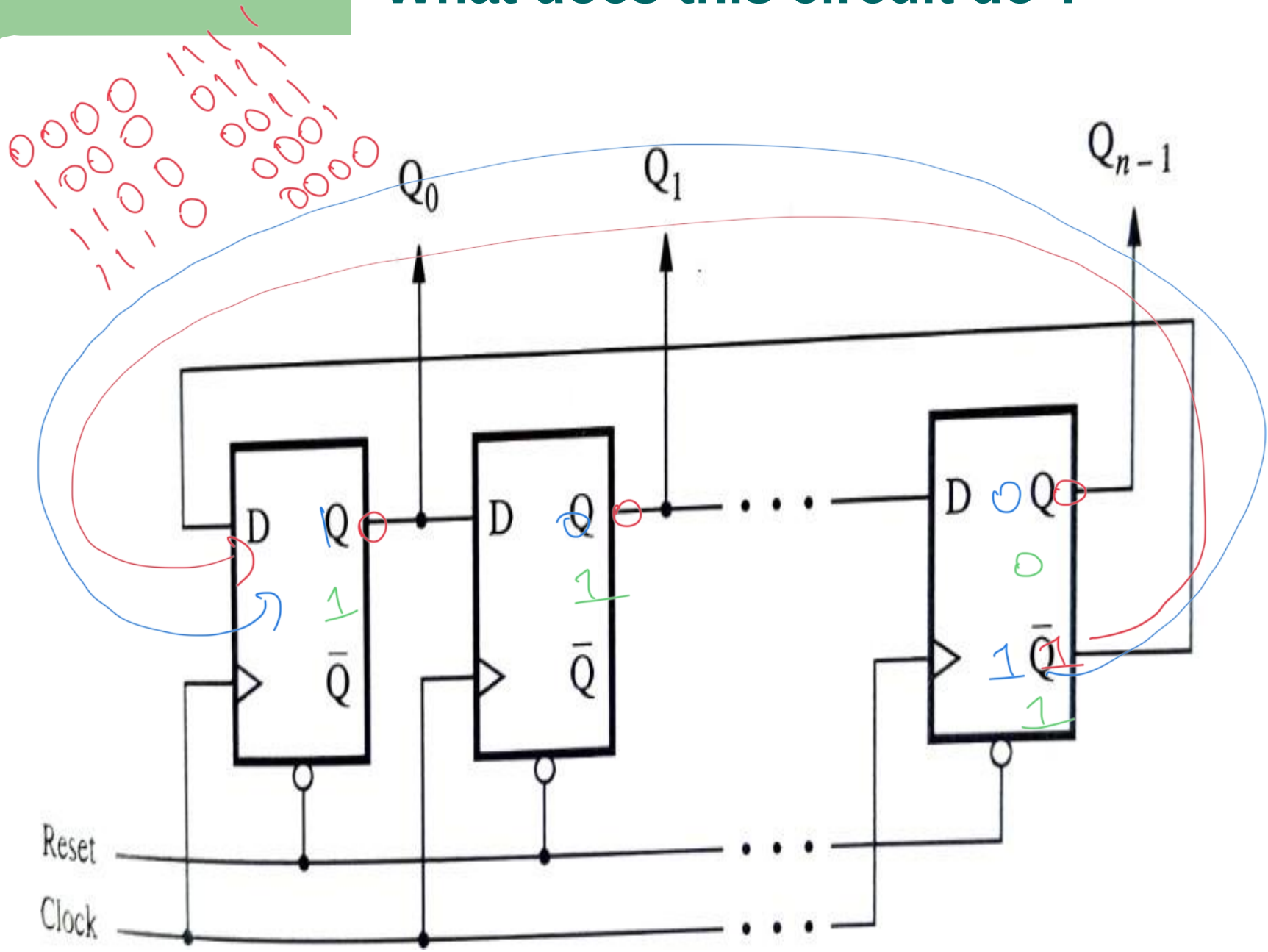
# Explanation

- The Start signal presets the leftmost flip-flop to 1 (since it is connected to preset pin) while it also clears the remaining flip-flops (to 0) since it is connected to the Clear pin.
- So if we had 4 flip-flops, we would have 1000 to begin with. Thereafter, in each clock cycle, we will have this 1 shifted to the right so we will have 0100, then 0010 and finally 0001.
- The circuit is called as a ring counter.

# Application of a ring counter

- A ring counter is used in hardware designs to create what are called finite-state machines.
- It uses what is called one-hot encoding (we have a single 1 and the rest of the code variables are 0).
- Essentially, we have one flip-flop per state. The finite state machine (that could correspond even to the control unit of a processor) moves from one state to another. Additional details in EE2016.

# What does this circuit do ?



# Student exercise

- Note the initialization via the reset signal.
- Identify the purpose of this circuit.

# Class Exercise

- Determine how the circuit behaves.

# Explanation

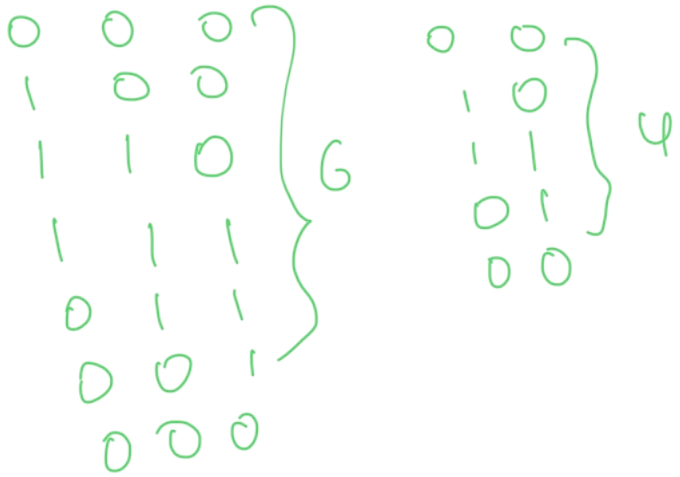
- The initialization for this is provided by the reset signal to all the flip-flops (so we have 0000 to begin with if we had 4 flip-flops).
- Thereafter, note that this circuit has the last (rightmost) flip-flop's Qbar output connected to the D input of the leftmost flip-flop.
- As a result, a 4 flip-flop circuit will count as 0000, 1000, 1100,1110,1111,0111,0011,0001,0000 ..
- It is called as a Johnson counter.

# Applications

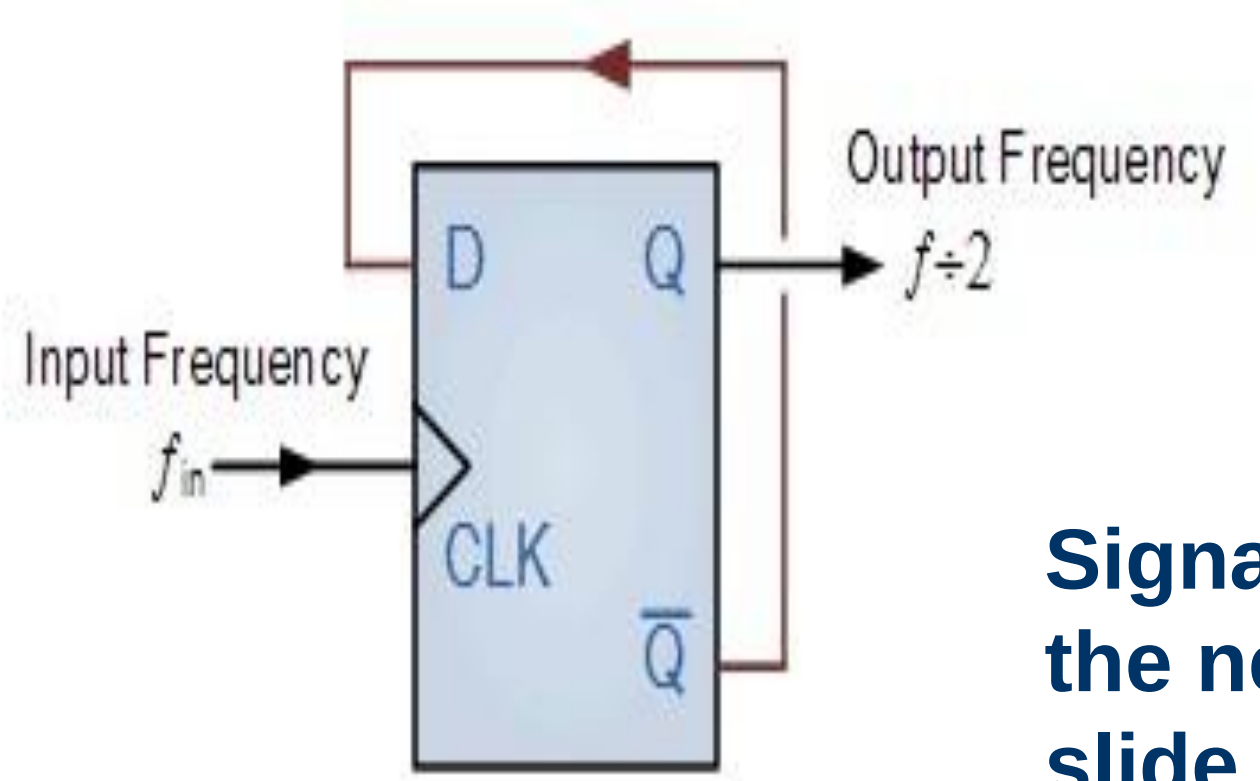
- It can be used in finite state machines just like a ring counter.
- Note that a 4-bit Johnson counter (with just 4 flip-flops) generates a counting sequence of length 8.

# Class Exercise

- Identify the counting sequence for a 3-bit Johnson counter.

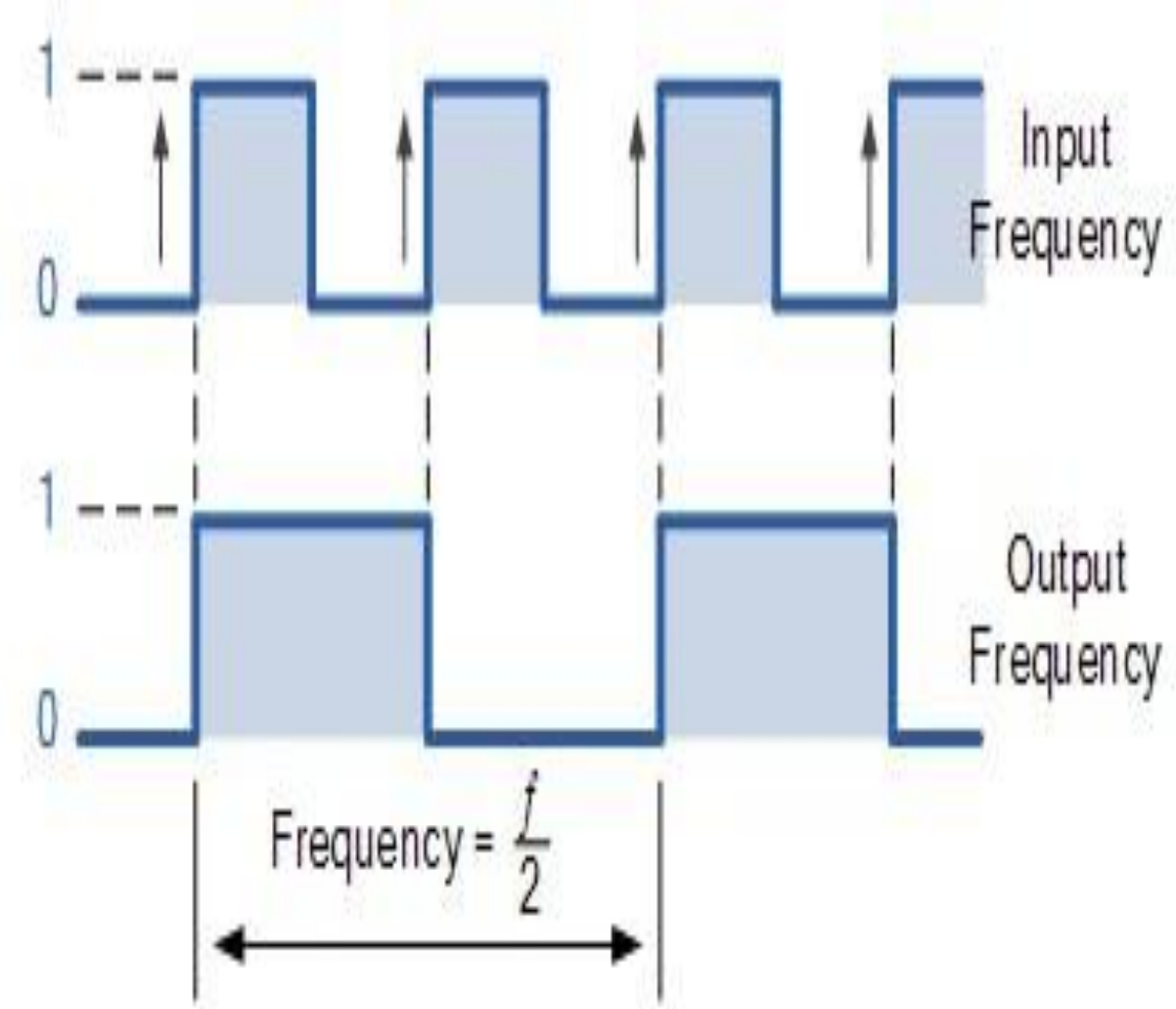


# Frequency division: Another application of flip-flops




**Signals in the next slide**

# Waveforms for the circuit shown



# Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.